# Exp: 3 Experiments on Cisco Packet Tracer

## Aim:

To understand the environment of Cisco Packet Tracer, design simple networks using a hub and a switch, and analyse the behaviour of the devices.

## Requirements:

* Cisco Packet Tracer
* PC with Windows / Linux

## Procedure (a) Hub network:

1. Open Packet Tracer and select End Devices; drag 4-6 PCs to the workspace.
2. Select Network Devices > Hubs and place one Hub.
3. Connect each PC to the hub using Copper Straight-Through cable (FastEthernet0 to hub port).
4. Assign IP addresses to the PCs (Desktop > IP Configuration), e.g. 192.168.1.1 to 192.168.1.5, mask 255.255.255.0.
5. Switch to Simulation mode, send a simple PDU from PC0 to PC3 and observe that the hub broadcasts the packet to all ports.

## Procedure (b) Switch network:

1. Replace the hub with a Switch (e.g. 2960) and connect the PCs with Copper Straight-Through cables.
2. Assign IP addresses in the same network.
3. In Simulation mode send a PDU from PC0 to PC3. The first (ARP) request is flooded, after which the switch learns MAC addresses and forwards frames only to the destination port.
4. Check the learned table in the switch CLI:

```
Switch> enable
Switch# show mac address-table
```

## Procedure (c) Hub and switch together:

1. Connect a hub and a switch with a Copper Cross-over cable and attach PCs to both.
2. Send PDUs and compare: the hub repeats a frame to every port, the switch forwards only to the destination port.

## Difference Between Hub and Switch:

* Hub: layer 1 device, broadcasts data to all ports, single collision domain, half duplex.
* Switch: layer 2 device, uses MAC address table to forward data to the ~~required~~ required port only, separate collision domain per port, full duplex.

## Network Topology of the College:

Observe the topology used in the college (e.g. star / extended star) and draw and label it here or in the observation book.

## Result:

Networks using a hub and a switch were designed in Cisco Packet Tracer and the difference in their behaviour was observed.